Comparative Performance of Traditional Machine Learning and Deep Learning architectures for Music Genre Classification using the GTZAN dataset

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Abstract

Music and numbers can be intertwined, contrary to prior belief that they are two separate entities given real life applications and imagination on utilizing such technical knowledge to something that is creative. This study investigates the comparative performance of traditional machine learning and deep learning architectures for music genre classification using the GTZAN dataset (10 genres, 999 tracks). The methodology employed a dual-feature approach: Static MFCC statistics for the Support Vector Machine (SVM) baseline, and Mel-Spectrograms for deep learning models, including a 2D Convolutional Neural Network (CNN), a Long Short-Term Memory (LSTM) network, and a custom Graph Neural Network (GNN) utilizing a temporal graph representation.

The results demonstrated a definitive hierarchy of performance, with the SVM Baseline achieving the highest accuracy of 73.00%. In contrast, the most effective neural network, the GNN, reached only 50.50%.

The SVM succeeded due to its robustness on small data and its reliance on highly effective, handcrafted features. Data scarcity fundamentally caused the limits to the performance ceiling for deep learning in this context. Future work must prioritize data augmentation and transfer learning to surpass the established 73.00% benchmark.

1 Introduction

In the generation where digitalization leads as we thread through the future, the exponential growth of audio libraries has made retrieving music information essential for organizing and discovering content. Music genre classification acts as the foundation of this field, but still remains a complex task due to the non-linear and subjective nature of musical styles. This study explores the tension between traditional feature-engineered models and deep-learning architectures by utilizing the GTZAN dataset (a benchmark containing 1,000 tracks across ten genres). While traditional methods are known for being robust on smaller data, they often lack in capturing the intricate, non-Euclidean “flow” inherent in musical compositions.

The primary problem addressed in this research is the data-scarcity bottleneck. Deep learning models, while theoretically powerful, are notoriously "data-hungry" and prone to overfitting when restricted to small-scale datasets like GTZAN. This creates a performance paradox where simpler models often maintain a higher performance floor than complex neural networks. Furthermore, standard spatial (CNN) and sequential (RNN) models treat audio as a flat grid or a linear string, potentially ignoring the relational dependencies—such as recurring motifs and harmonic transitions—that truly define a genre's identity.

The hypothesis of this study is that by modeling music as a relational graph rather than a linear sequence, a Graph Convolutional Network (GCN) can capture stylistic "flow" more effectively than standalone neural architectures. Specifically, it is hypothesized that the relational inductive bias of a GCN will provide a more sophisticated framework for identifying nuanced genre patterns, though its generalization ability may be critically tested by the limited sample size of the dataset. This research aims to benchmark this graph-based approach against a Support Vector Machine (SVM) baseline to map the threshold where structural complexity meets data reality.

2 Related Work

As creativity expands in the minds of the people, music has also been growing diverse as outputs of how humans have been expressing themselves to the world. To further understand the message of what the people would want to portray, music genre has been a way to

simplify it into classifications. Many recent studies have utilized machine learning techniques in different approaches to classify genres of various music.

To demonstrate machine learning on music genre classification, Prajwal et.al. (2021) used Librosa library on the GTZAN dataset for audio analysis and extracted a set of features such as zero-crossing rate (measure of rate of sign changes along the signal [positive to negative or back], Spectral Centroid (weighted mean of the frequencies in the sound and locates the center of mass of the sound), Spectral Rolloff (measures the shape of the signal), Mel Frequency Capstrum Coefficients (MFCC's) – set of features that descrive the overallshape of the spectral envelope), and Chroma Feature. Among the different models tested, Support Vector Machine classifier showed the highest accuracy at 73.2% (without hyper-parameters tuning) and 76.4% (after hyper-parameters tuning).

Additionally, Akhtar and Mishra (2020) made a comparative study using the GTZAN dataset and revealed that the Support Vector Machine (SVM) classifier achieved the highest classification performance. Their study specifically emphasized that with the strong bias from feature engineering, achieves lower variance and better generalization on this specific dataset compared to the high-variance, low-bias Convolutional Neural Network (CNN) model. They explicitly attribute this to the "moderately sized" (small) nature of the GTZAN dataset, where CNNs struggle to generalize compared to the strong inductive bias of handcrafted features. While deep learning is often considered superior for large-scale tasks, their research revealed that for corpora with limited samples (~100 per class), this hierarchy is effectively inverted. They posit that high-capacity models (CNNs/LSTMs) suffer from high variance and overfitting, whereas traditional classifiers like the SVM maintain robustness due to their strong inductive bias. This study seeks to experimentally test this inversion by benchmarking a traditional SVM against three distinct neural architectures.

Furthermore, several model architectures were also tested using the GTZAN Dataset and the Ballroom dataset, which was also a genre classification dataset by Ahmed et. al. (2024). This study discovered that across the model architectures: Convolutional Neural Network, Feedforwarding Neural Network, Support Vector Machine, k Nearest Neighbors, and LSTM, the modified CNN model reached the highest accuracy at 92.7%. This was followed by CNN, SVM, FNN, RNN-LSTM, and k-NN respectively.

While spatial and sequential models have dominated the field, recent research had shifted toward Geometric Deep Learning to better represent the inherent structure of audio. Nowak and Zieliński (2025) demonstrated that representing music as a graph allowed a more nuanced capture of 'relational flow' than traditional Euclidean grids. This graph defined that nodes (which represented discrete musical events such as notes, chords, or measures) encapsulate temporal segments and edges encode temporal flow, relational flow, and stylistic transitions. Adopting this rationale to the study, implementing a Graph Convolutional Network (GCN) to evaluate whether this graph-based inductive bias can overcome the generalization hurdles identified in standalone CNN and LSTM architectures, particularly when dealing with the non-linear transitions inherent in musical genres like jazz and classical music that share similar notes but have entirely different harmonic structures or rhythmic phrasing. This exact rationale led to an average accuracy of 89.3% on GCN, significantly outperforming traditional CNN (75.1%) and RNN (79.4%) baselines.

Recent advancements in audio analysis have shifted toward representing sound as a graph to capture inherent dependencies. Castro-Ospina et al. (2024) demonstrated that GNN-based approaches, particularly Graph Attention Networks (GAT) and Graph Convolutional Networks (GCN), are highly competitive, achieving up to 91% accuracy in environmental sound classification. Their methodology highlights a shift from treating audio frames as independent points to modeling them as connected nodes, a framework to be adopted to explore music genre relationships.

By synthesizing the robust baseline established by Prajwal et al. with the relational innovations of Nowak and Zieliński, this project evaluates the threshold at which handcrafted feature engineering meets its limit. By implementing a GCN architecture, this study aims to determine if the 'relational flow' of musical segments can mitigate the data-scarcity challenges that typically hinder standalone deep learning models on the GTZAN dataset.

3 Model

3.1 Support Vector Machine

The Support Vector Machine served as the traditional machine learning baseline model of the study. It was chosen for its known effectiveness on small-to-medium datasets when paired with domain-specific features (Prajwal et al., 2021). The input was a 48-dimensional Static Mel-Frequency Cepstral Coefficient (MFCC) vector. Utilizing a Radial Basis Function (RBF) kernel, the model mapped features into a high-dimensional space to maximize the margin between the ten genre classes—an architecture specifically noted for its robustness against the overfitting challenges typical of deep learning on the GTZAN corpus (Mishra & Akhtar, 2025).

The following three neural network architectures were implemented to process the Mel-Spectrogram input, with each one specialized for a different dimension of the data:

3.2 2D Convolutional Neural Network (CNN)

Convolutional Neural Network was implemented to treat the Mel-Spectrogram as a 2D image, leveraging the model's proficiency in identifying local spatial correlations within the time-frequency plane (Ahmed et al., 2024). CNNs are proficient at discovering local spatial correlations. In a Spectogram, this is parallel to recognizing patterns of energy that occur across adjacent frequency bins and short periods of time (e.g. the shape of a drum hit or a specific harmonic cluster). This approach enables the recognition of spectral "textures"—patterns of energy occurring across adjacent frequency bins and short temporal intervals, such as rhythmic strikes or harmonic clusters (Ashraf et al., 2023). In this study, multiple layers of 2D Convolution followed by MaxPooling were used as the architecture of the model to progressively minimize the spatial dimensions while obtaining increasingly complex features, culminating in Dense layers for classification.

3.3 Long Short-Term Memory Network

In the previous model, the input feature was treated as an image. Meanwhile, the Long Short-Term Memory Network that is a variant of Recurrent Neural Network (RNN) was implemented to treat the Mel-Spectogram as a temporal sequence. This model is leading at modeling long-term dependencies. In audio, this means linking events that are far apart in the music (e.g., the relationship between a verse and a chorus) (Ashraf et al., 2023).

In this study, this model utilized two consecutive LSTM layers followed by a Dropout layer to mitigate overfitting, concluding with a Time Distributed Dense layer before final classification.

3.4 Custom Graph Neural Network

The GNN (specifically a Graph Convolutional Network or GCN) was implemented as an advanced experimental architecture to model the music's relational flow (Nowak & Zieliński, 2025). This study approaches the data by transforming the audio into a temporal graph, where each node represented a 1-second segment of the song, and edges linked chronologically adjacent segments. GNN was chosen to explicitly model the local and global connections between these musical segments, allowing information to flow across the song's structure in a non-linear way (Castro-Ospina et al., 2024).

In this experiment, two layers of GCNConv were implemented to aggregate information across the temporally connected nodes, with a Global Mean Pooling layer used as the readout function to convert the final node features into a single, fixed-size graph embedding for classification.

3.5 Dense Head Classifier (Simulated Transfer Learning)

This experimental setup was designed to test the quality of the features learned by the GNN. If the GNN had learned a truly "substantial" representation of music, a simple classification head should be able to achieve high accuracy using those features. In this study, the final classification layer was removed from the pre-trained GNN. These 64-dimensional vectors were then fed into a separate sequential Dense Head—comprising Linear and ReLU layers—which was trained independently. This methodology isolates the GCN's representational power, a technique used in state-of-the-art research to measure the effectiveness of node information extraction on downstream tasks (Castro-Ospina et al., 2024).

4 Experiments

4.1 Datasets

This project utilizes the widely referenced GTZAN Music Genre Classification Dataset (Tzanetakis & Cook, 2002), accessed via the curated distribution provided by Olteanu (2019). The dataset consists of 1,000 thirty-second audio tracks evenly distributed across 10 musical genres (100 tracks per genre). After initial filtering, a total of 999 tracks were used for modeling. The data was partitioned using an 80/20 stratified split, yielding $N_{train} = 799$ tracks for training and $N_{test} = 200$ tracks for testing across all models to ensure robust comparison.

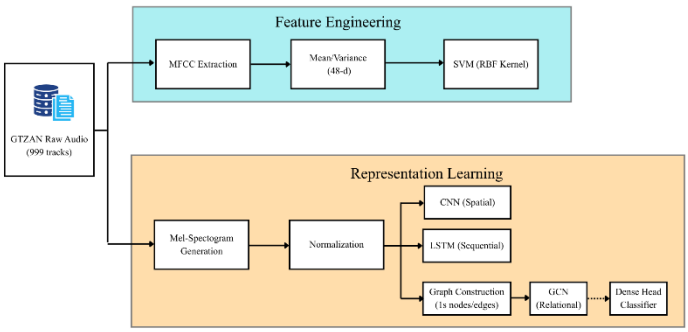

Figure 1. Proposed Multi-Stream Framework for Music Genre Classification. The architecture illustrates the bifurcation of the GTZAN dataset into two analytical pipelines: (A) the Traditional Stream, utilizing handcrafted 48-dimensional MFCC vectors for the SVM baseline, and (B) the Deep Learning Stream, utilizing normalized Mel-spectrograms. Stream B evaluates three distinct architectural inductive biases: spatial (CNN), sequential (LSTM), and relational (GCN), with the latter incorporating a custom graph construction phase. Additionally, a Dense Head Classifier is utilized for linear probing to diagnose the quality of the latent embeddings generated by the GCN.

4.2 Feature Engineering and Representation

Two distinct feature representations were extracted from the raw audio for comparative analysis on traditional and deep learning approaches, following the multi-stream framework illustrated in Figure 1:

4.2.1 Static Feature Vector (for SVM):

A 48-dimensional vector was constructed by calculating the mean and variance of the first 24 Mel-Frequency Cepstral Coefficients (MFCCs) across each 30-second window. This compact representation captures the global "timbral fingerprint" of the audio, providing a robust baseline for the non-probabilistic linear classifier.

4.2.2 Sequential/Image Representation (for Neural Networks):

For the CNN, LSTM, and GCN models, raw audio was converted into Mel-spectrograms—a 2D visual representation of power across frequency and time. All spectrograms were standardized using Z-score normalization to stabilize gradient descent during the training of the neural architectures.

4.2.3 Representation Probing (Dense Head Classifier)

To evaluate the quality of the features learned by the GCN, a simulated transfer learning experiment was conducted. This process involved extracting the 64-dimensional latent embeddings from the pre-trained GCN by removing its final classification layer, as shown by the dashed diagnostic path in Figure 1. These embeddings were then fed into a standalone Dense Head classifier—a sequential architecture consisting of Linear layers and Rectified Linear Unit (ReLU) activation functions. By freezing the GCN's weights and training only the new dense layers, the study isolated the model's representational power. This diagnostic step determines if the GCN captured a "substantial" understanding of musical structure that is robust enough for downstream tasks without further architectural updates.

4.3 Implementation Details

The experimental framework was developed using Python 3.10, leveraging a diverse, research-grade software ecosystem to handle both traditional and geometric data structures. Signal processing and feature extraction, including the generation of high-fidelity MFCCs and Mel-spectrograms, were conducted using the Librosa library. Traditional machine learning benchmarks were implemented via scikit-learn, while the spatial and sequential neural architectures (CNN and LSTM) were built using the TensorFlow/Keras framework. To facilitate the non-Euclidean modeling required for the custom GNN, PyTorch Geometric (PyG) was utilized for its specialized graph-handling capabilities. All neural

models were trained using the Adam optimizer and categorical cross-entropy loss, with a consistent dropout rate of 0.3 applied to the dense layers to balance model depth with the generalization requirements of a limited training set.

5 Results and Discussion

Final Model Performance Metrics

The performance of the four primary architectures and the supplemental transfer learning simulation were evaluated based on Test Set Accuracy on the held-out 200 tracks (20% of the dataset).

The experimental evaluation produced a decisive outcome that characterizes the relationship between model complexity and data scale. As shown in Table 1, the Support Vector Machine (SVM) established the highest performance benchmark, achieving an accuracy of 73.00%, which significantly outperformed the best-performing neural architecture, the Graph Convolutional Network (GCN), at 50.50%. This 22.5 percentage point gap directs the focus of the analysis toward the project's central constraint: the data-scarcity bottleneck inherent in the GTZAN corpus.

Table 1, Final Test Accuracies of the Model Architectures and their Input Features.

Rank	Model Architecture	Input Features	Final Test Accuracy
1	**Support Vector Machine (SVM)**	Static MFCC Statistics (48 features)	**73.00%**
2	**Graph Neural Network (GNN)**	Segmented Mel-Spectrograms (Temporal Graph)	**50.50%**
3	**Tuned Long Short-Term Memory (LSTM)**	Mel-Spectrograms (Sequential)	**48.50%**
4	Dense Head Classifier (Simulated Transfer Learning)	GNN Extracted Features (64-dim embedding)	**44.50%**
5	2D Convolutional Neural Network (CNN)	Mel-Spectrograms (Image)	**42.50%**

The superior performance of the SVM baseline validates the first research objective, confirming that domain-specific feature engineering provides a critical "safety net" in small-data environments. As presented in Figure 2, the SVM achieved notable recall for Jazz (19/20) and Classical music (18/20). By utilizing a compact, 48-dimensional MFCC feature vector, the SVM effectively captured discriminative timbral qualities without the high risk of overfitting associated with millions of trainable parameters. However, the matrix also highlights a critical limitation: the model frequently misclassifies Reggae as Hiphop and Rock as Pop. This "Timbral Overlap" indicates that while MFCCs are robust, distinguishing between genres that share similar instrumentation but differ in rhythmic "flow" is a challenge. This outcome confirms the "inverted hierarchy" identified by Mishra and Akhtar (2025), where traditional classifiers trained on handcrafted features maintain higher robustness than high-capacity deep learning models when limited to fewer than 100 samples per class.

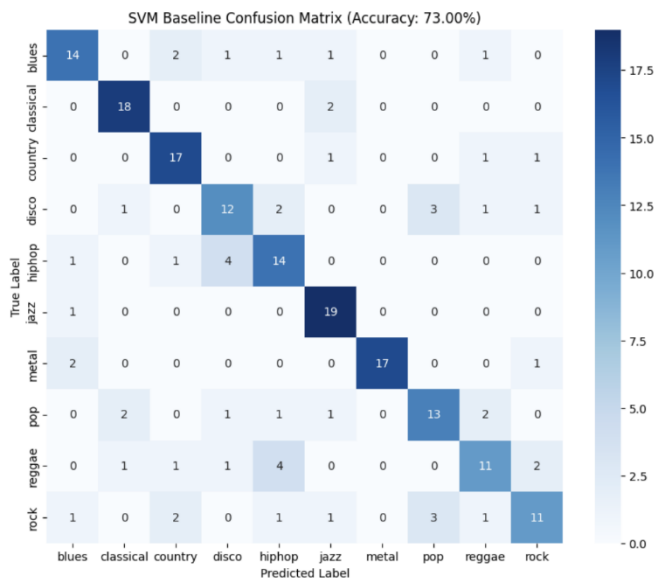

Figure 2. Confusion Matrix of the Support Vector Machine Baseline (73.00%)

While the SVM excelled in timbral identification, the neural architectures were implemented to test the study's hypothesis regarding relational modeling. Although all deep learning models were constrained by the 999-song dataset, the GCN emerged as the most effective neural paradigm, outperforming both the spatial CNN and sequential LSTM baselines. This performance margin suggests that modeling music as a non-Euclidean relational graph—where 1-second segments are connected by temporal edges—is more effective for capturing the inherent "flow" of a composition than treating audio as a standard image or a simple linear sequence. This finding supports the

rationale provided by Nowak and Zieliński (2025) regarding the superiority of graph-based inductive biases in music information retrieval.

To further diagnose the quality of the neural representations, a linear probing evaluation was conducted using a standalone Dense Head Classifier. The resulting accuracy of 44.50%—a drop from the end-to-end GCN performance—revealed that the internal features generated by the GCN were not yet sufficiently generalized for effective transfer learning. This performance drop proves that in the absence of large-scale pre-training, the model struggles to extract high-dimensional features that are "substantial" enough to be utilized in a frozen-weight context.

Ultimately, this study demonstrates that while architectural complexity (moving from CNN→LSTM →GCN) provides marginal gains in relational modeling, these improvements remain secondary to the influence of data scale. The results confirm that to bridge the gap between handcrafted robustness and neural intelligence, future work must prioritize the integration of pre-trained embeddings (such as VGGish or YAMNet) to initialize graph nodes. By addressing the "Extraction Challenge" identified in recent literature, the structural benefits of the GCN can be fully realized, potentially surpassing the ceiling established by traditional machine learning baselines. Furthermore, to achieve modern classification accuracy, the project must shift its focus from model tuning to data expansion.

6 Conclusion and Recommendations

This study investigated music genre classification on the GTZAN dataset, employing both traditional machine learning (SVM) and state-of-the-art deep learning models (CNN, LSTM, GNN).

The definitive finding of this research is the superiority of the Support Vector Machine (SVM) Baseline, which achieved a final test accuracy of 73.00%. This performance benchmark significantly exceeded all tested deep neural network architectures, including the Graph Neural Network (GNN), which was the highest-performing neural architecture at 50.50%.

The findings suggest that the primary performance constraint is the limited size of the GTZAN dataset (999 tracks). This data scarcity fundamentally limited the ability of the high-capacity deep learning models, specifically the LSTM and CNN to generalize, resulting in a consistent performance ceiling around 50%. Furthermore, the Dense Head Classifier experiment with the accuracy at 44.50% confirmed that internal representations learned by GCN were not yet sufficiently robust for transfer learning, which gives emphasis on the conclusion that model simplicity and feature engineering currently outweigh architectural complexity in data-constrained environments. This explains the SVM's success was rooted in its robustness to small data and its reliance on effective, handcrafted feature engineering (MFCC statistics).

In essence, the project demonstrates that in data-constrained environments, model simplicity and feature quality are more critical than architectural complexity. To surpass the 73.00% barrier, future work must abandon standalone architectural tuning and shift toward a multi-layered strategy focused on data expansion and architectural hybridization. The primary priority remains overcoming data scarcity through audio augmentation (pitch-shifting and time-stretching) together with the development of Graph-CNN hybrids that combine spatial feature extraction with relational temporal modeling.

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